

Manufacturing Operations Analytics

Executive Summary | May 6, 2026

Overview

This analysis examines production output, defect rates, and downtime incidents across five manufacturing factories using ERP export data. The objective is to identify underperforming factories, determine the leading causes of defects, and pinpoint where production time is being lost. Data was cleaned and standardized in Python before analysis and visualization in Tableau.

Data

File	Rows (Raw)	Rows (Cleaned)
production_log.csv	1,860	1,578
defect_log.csv	1,245	1,078
downtime_log.csv	935	696

Data quality issues resolved: mixed date formats (4 variations), inconsistent factory name conventions, duplicate rows, numeric values stored as text, impossible values (negative production output, downtime exceeding 24 hours), and missing values in critical columns.

Key Findings

1. Factory D is the clear underperformer across all metrics.

Factory	Total Units	Avg Daily Units	Defect Rate	Downtime Hours
Factory C	79,369	256.0	16.6%	448.9
Factory A	79,135	252.8	5.6%	442.0
Factory B	75,649	257.3	5.8%	548.7
Factory E	60,500	193.3	6.0%	430.4
Factory D	35,562	102.2	17.3%	1,347.4

Factory D produced 35,562 total units at an average of 102 per day — less than half the output of Factories A, B, and C. Its 17.3% defect rate is nearly three times the 5.6–6.0% range of the remaining factories. It also accumulated 1,347 downtime hours, more than double any other factory.

2. Material Defect and Assembly Error are the leading defect causes.

Defect Reason	Total Defects
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Material Defect	3,564
Assembly Error	3,344
Machine Malfunction	2,077
Supplier Issue	1,903
Human Error	1,676
Quality Control Failure	1,430

Material Defect and Assembly Error together account for nearly 47% of all defects. These are distinct root causes — supplier quality issues versus assembly line process failures — and require separate interventions.

3. Equipment Failure is the leading unplanned downtime cause.

Cause	Total Hours	Incidents
Equipment Failure	690.0	155
Scheduled Maintenance	680.9	153
Raw Material Shortage	512.6	104
Safety Incident	456.0	106
Power Outage	442.7	88
Staff Shortage	435.2	90

Equipment Failure is the top unplanned downtime cause at 690 hours across 155 incidents. Scheduled Maintenance at 681 hours is expected and planned. The priority is reducing unplanned failures through preventive maintenance.

Recommendations

1. Audit Factory D before increasing production targets. Factory D's 17.3% defect rate means a significant portion of its output is defective. Increasing production volume without resolving the quality problem will generate more defective units, not more usable output. Investigate root causes of both high defects and high downtime before setting new targets.

2. Address Material Defect and Assembly Error as separate problems. Material Defect points to supplier quality — review supplier contracts and incoming material inspection processes. Assembly Error points to process or training failures on the line — review assembly procedures and operator training programs.

3. Implement a preventive maintenance schedule to reduce Equipment Failure. At 690 hours and 155 incidents, Equipment Failure is the top unplanned downtime cause. A preventive maintenance program shifts downtime from unplanned to planned, reducing production disruption.

4. Flag Factory C as a secondary quality priority. Factory C has the highest total production output but a 16.6% defect rate — nearly as high as Factory D. High output combined with high defect rate means Factory C is producing a large volume of defective units. This warrants investigation after Factory D is addressed.